

EU AI Act Risk Classification Tool

Technical README and Project Documentation

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GitHub: github.com/brandonmuch
Source Legislation: EUR-Lex 2024/1689, EU Artificial Intelligence Act
Language: Python 3
File: eu_ai_act_classifier.ipynb

Overview

The EU AI Act Risk Classification Tool is a Python-based compliance instrument that helps organisations determine their legal obligations before deploying an AI system in the European Union.

Given a plain-language description of an AI system, the tool returns:

- The applicable risk tier under EU Regulation 2024/1689
- The specific legislative article governing that classification
- The compliance obligations the deploying organisation must meet
- The fundamental rights principle the classification protects

Built directly from primary legislation, not third-party summaries or interpretive documents.

Risk Tiers

Tier	Article	Description
Unacceptable Risk	Article 5	Prohibited entirely. Cannot be deployed in the EU.
High Risk	Article 6 + Annex III	Permitted with strict compliance obligations.
Limited Risk	Article 50	Permitted with transparency obligations only.
Minimal Risk	Recital 48	Permitted. No mandatory obligations.

Legislative Basis

This tool maps directly to the following provisions of EU Regulation 2024/1689:

Article 5: Prohibited AI Practices

Systems that manipulate human behaviour, enable social scoring, perform real-time biometric surveillance in public spaces, infer sensitive personal characteristics from biometric data, or predict criminality from personality traits are prohibited entirely under the Act.

Article 6 and Annex III: High-Risk AI Systems

Systems deployed in critical infrastructure, employment decisions, credit and financial services, education assessment, law enforcement, border control, medical devices, and biometric identification are subject to mandatory conformity assessments, risk management systems, human oversight requirements, and EU database registration.

Article 50: Transparency Obligations

Chatbots, deepfakes, synthetic media, and emotion recognition systems must clearly disclose to users that they are interacting with an AI system.

Recital 48: Minimal Risk

AI systems with negligible risk to fundamental rights or public safety, such as spam filters and recommendation engines, carry no mandatory obligations under the Act.

Core Principle

The EU AI Act is unified by a single philosophical foundation. AI systems must be human-centric, non-discriminatory, and dignity-preserving. Every prohibition and obligation in the Act flows from this principle.

How to Use

Clone the repository and open eu_ai_act_classifier.ipynb in Jupyter Notebook.

Run all cells in order:

- Cell 1: Header and confirmation
- Cell 2: Tier definitions
- Cell 3: Keyword mappings and classifier function
- Cell 4: Test function and sample cases

To test your own AI use case, add a new line at the bottom of Cell 4:

```
test("Describe your AI system here in plain language")
```

Example Outputs

Input: An AI that screens CVs for job applications

Classification: HIGH RISK, Article 6 + Annex III

Obligations: Requires conformity assessment, risk management system, human oversight, and EU registration.

Input: A customer service chatbot

Classification: LIMITED RISK, Article 50

Obligations: Must disclose to users that they are interacting with an AI system.

Input: A system that scores citizens based on social behaviour

Classification: UNACCEPTABLE RISK, Article 5

Obligations: Cannot be placed on the market or put into service in the EU.

Known Limitations

This tool uses keyword matching against plain-language descriptions. Classification accuracy depends on the specificity and clarity of the input description. Edge cases and hybrid AI systems may require manual review against the full legislative text.

This tool is a governance aid, not a legal instrument. All outputs are indicative only.

Disclaimer

This tool is indicative only. Final risk determination for any AI system requires qualified legal review under EUR-Lex 2024/1689. The author accepts no liability for compliance decisions made on the basis of this tool's output alone.

Author Background

Built by Brandon Muchenje, an AI Governance and GRC Analyst with a BCom in Law (Cum Laude) and training across the EU AI Act, NIST AI RMF, ISO/IEC 42001, GDPR, and GRC fundamentals. Targeting AI governance and regulatory compliance roles in the Netherlands, Luxembourg, and UK.

References

- EU AI Act Full Text: eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32024R1689
- NIST AI Risk Management Framework: nist.gov/artificial-intelligence
- ISO/IEC 42001, AI Management Systems: iso.org/standard/81230.html